Unsupervised Learning Laboratories

Lab 4

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4 Kernel PCA

In Lab 2 we saw that PCA, being a linear projection, cannot separate data whose classes are not linearly separable. Kernel PCA performs PCA in an implicit feature space defined by a kernel, allowing nonlinear structure to be captured. You can use external libraries for linear algebra operations (e.g. `numpy.linalg`) but you are expected to write your own algorithms.

4.1 Exercise 1

Use the files `kPCA_data_2024.txt` and `kPCA_labels_2024.txt` in the `Datasets` folder. The first file contains the variables describing the data, the second one the labels of the classes associated to it.

- Visualize and explore the data: how many dimensions does it have? What do you see if you plot the first two dimensions?
- Apply your own implementation of PCA to the dataset and plot the eigenvalue spectrum.
- Project the data on the first two principal components and colour by class.
- Implement your own version of Kernel PCA.
- Apply Kernel PCA to the dataset. Test both a Gaussian kernel with width $\sigma \in [0.05, 2.0]$ and a polynomial kernel varying the degree $\delta \in \mathbb{N}$.
- Plot the transformed data in 2D and 3D for the different kernels, and comment on which kernel and parameters separate the classes best.

Notes

- Use the second version of the polynomial kernel introduced in class (the one with "$1 + \ldots$").
- Remember to centre the kernel matrix.